Identification of genetic biomarkers associated with overall survival across treatment mechanisms of action in metastatic colorectal cancer patients

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Abstract

Colorectal cancer (CRC) is the second most fatal cancer in the US, with metastatic CRC carrying a 5-year survival rate of 13%. Identifying genetic biomarkers that are associated with overall survival (OS) across treatment groups could improve patient outcomes. Using the MSK-CHORD database, 1,305 metastatic CRC patients were analyzed to assess whether the top 10 most frequently mutated genes in this cohort were associated with OS across 3 treatment mechanisms of action (MoA): chemotherapy, chemotherapy with targeted agent, and other/ investigational. Multivariable Cox proportional hazards models with gene x MoA interaction terms made up the primary analysis, which was supplemented by a landmark analysis, RMST, Weibull AFT models, bootstrap validation, and sensitivity analyses. In the chemotherapy group, BRAF, SMAD4, PIK3CA, and KRAS mutations were associated with a harmful OS, while APC was associated with a protective OS. No interaction survived FDR correction. In secondary analyses, robustness of results was confirmed, and the PIK3CA x chemotherapy and targeted in the late period showed potential significant interaction, further supported by AFT and bootstrapping. These results highlight the relevance of mutations in metastatic CRC and identify PIK3CA as a candidate biomarker for treatment-specific effects needing further investigation.

Introduction

Cancer is the number two leading cause of death in the US, and colorectal cancer (CRC) is the second most fatal cancer in the US. It is estimated that approximately 158,850 new cases will be diagnosed and 55,230 people will die from CRC in this year alone.[1] However, with early treatment, 5-year survival rates for localized CRC is around 90%. There is a sharp drop in survival as CRC progresses further to metastatic stages, with a 5-year survival rate of around 13%.[2] The treatment landscape for metastatic CRC is vast, including chemotherapy, targeted biological agents, immune checkpoint inhibitors, and investigational therapies. However, despite this breadth of options, treatment recommendations follow traditional strategies and individual patient outcomes vary considerably.

As genetic data becomes increasingly accessible, there sits a very interesting question at the intersection of cancer genomics and pharmaceutical sciences: are there genetic biomarkers that can explain differentiated treatment responses? To investigate this question, data from Memorial Sloan Kettering Cancer Center Clinicogenomic, Harmonized Oncologic Real-World data set (MSK-CHORD) was analyzed, which is a large real-world oncology dataset that links tumor genetic profiles to longitudinal clinical outcomes and treatments.[3]

Methods

Data and Study Population

Data came from the MSK-CHORD database, from which 6 data files were used: clinical patient data (demographics, OS status, OS time, highest cancer stage, microsatellite instability (MSI) score, and tumor mutational burden (TMB)), timeline diagnosis (diagnosis dates and diagnosis information), timeline treatments (agents, dates of agents), and mutation data (Hugo symbol).

The analytic cohort was restricted to only metastatic CRC patients. From 5,543 total CRC patients in MSK-CHORD, there were 1,448 metastatic CRC patients. These patients were then assigned a first-line treatment regimen, which represented the MoA of the first treatment started on or after the diagnosis date for metastatic CRC. The top ten most frequently mutated genes were then identified. Missing OS time or OS status were dropped.

Variables

The primary outcome was OS, defined as the time in months from the start of first-line treatment until death. Patients who were alive at the last follow-up were right censored.

The treatment groups were classified into 3 main groups: cytotoxic chemotherapy (n = 1,157), chemotherapy and targeted agent (n = 76), and other/ investigational (n = 85). Cytotoxic chemotherapy included agents such as capecitabine, oxaliplatin, romidepsin, etc. Chemotherapy and targeted agents included a cytotoxic chemotherapy agent and a VEGF inhibitor (i.e. bevacizumab) or EGFR inhibitor (i.e. cetuximab or panitumumab). The other/ investigational agents included PDL1 inhibitors, durvalumab, epcoritamab, and others. A consolidated MoA variable served as the indicator for MoA group, with chemotherapy as the reference in all models.

The top ten most frequently mutated genes were identified and represented as a binary variable indicating presence (mutant) or absence (wildtype) of a somatic mutation. Covariates included age at diagnosis (continuous), age squared, sex, race, highest stage recorded of cancer, log-transformed MSI score, logMSI squared, log-transformed TMB, and logTMB squared.

Primary Analysis: Multivariable Cox Proportional Hazards (PH) Models

For all ten genes, a separate multivariable Cox PH model was fit, which included a binary mutation indicator, the MoA group, an interaction term between those two, and all clinical covariates mentioned above. The general model specification was:

$$Surv(OS_{time}, OS_{event}) \sim gene_{mutation} + MoA + gene_{mutation} \times MoA + covariates$$

Hazard ratios (HR) and 95% confidence intervals were estimated for all model terms, with the interaction term being the primary parameters of interest as they represent whether the association between mutation and OS differs across MoA group. A total of 20 interaction terms were tested across all 10 models. To account for multiple comparisons, p-values were adjusted using the false discovery rate (FDR) Benjamini-Hochberg method. Significance was assessed at an uncorrected threshold of $p < 0.05$, and the FDR-adjusted threshold of $p < 0.05$.

Proportional Hazards Assumption Testing

The PH assumption was tested using Schoenfeld residuals, with a statistically significant global test indicating violation of this assumption. Upon discovery of the violation, secondary analyses were performed.

Landmark Analysis

To assess whether interaction effects differed across the course of the disease, a landmark analysis was performed. The cohort was split at median follow-up time of 35.5 months into an early and late period. Separate Cox models with the interaction terms were fit within each period for specific genes that

had PH violations (i.e. APC, PIK3CA, BRAF). FDR correction was then applied to p-values of interaction terms for each landmark period.

Restricted Mean Survival Time (RMST) and Accelerated Failure Time Models (AFT)

RMST analysis was conducted for APC, BRAF and PIK3CA as an alternative to the Cox model upon discovery of PH assumption violations. RMST estimates the average survival time up to a pre-specified time threshold. This time threshold was decided by taking the minimum survival time by gene x MoA group, and taking 90% of that time. Differences in RMST between mutant and wildtype patients were estimated within each MoA group, with 95% confidence intervals.

Weibull AFT models were fit for APC, BRAF, and PIK3CA interactions as another robustness check under PH violations. AFT was selected to model the multiplicative effect of covariates on OS instead of the hazard rate. HR equivalents were derived from AFT estimates as $e^{-coefficient}$, to be comparable to Cox model estimates. To assess robustness of interaction findings, directional agreement between Cox and AFT results was evaluated.

Bootstrap Validation

For the interaction hazard ratio estimates from the Cox models, 250 bootstrap resamples of the full data set were drawn with replacement. The Cox models were refit on each resample for the genes identified from uncorrected Cox Model significance testing (BRAF, APC, PIK3CA, and SMAD4). Bootstrapped 95% confidence intervals were constructed from the 2.5 and 97.5 percentiles of the distribution of HR interaction estimates across all 250 resamples. If the bootstrapped confidence interval did not contain 1, it was considered robust.

Sensitivity Analysis

Deviance residuals were computed from each Cox model in order to identify influential observations. Patients with an absolute deviance residual greater than 3 were treated as outliers. 7 patients (0.54%) were flagged as outliers. Sensitivity analyses refitting the Cox models after removing these 7 outliers and subsequent interaction HR estimates were calculated. The percent change between the original HR estimates and the sensitivity analysis's HR estimates were used to assess the influence of these 7 outliers on results.

Statistical Software

All analyses were conducted in RStudio version 2026.01.1+403. Key packages include: survival, survRM2, boot, and ggsurvplot.

Results

Study Population

The final analytic cohort consisted of 1,305 unique metastatic CRC patients. The majority of patients received cytotoxic chemotherapy (n = 1,157, 88.7%), then other/ investigational (n = 85, 6.5%), and finally chemotherapy with targeted agent (n = 76, 5.8%). The top ten most frequently mutated genes in descending order were: TP53 (78.9%), APC (76.6%), KRAS (45.5%), SMAD4 (16.8%), PIK3CA (15.9%), TCF7L2 (10.6%), FBXW7 (9.0%), SOX9 (8.6%), BRAF (7.7%), and ARID1A(6.4%). These genes encompass many different oncogenic pathways, including WNT signaling, RAS/MAPK signaling, PI3K/AKT signaling, and cell cycle regulation.

Table 1. Baseline Characteristics by First-Line Treatment Group

Characteristic	Overall N = 1,305[1]	Chemo N = 1,145[1]	Chemo+Targeted N = 75[1]	Other/IO N = 85[1]	p-value[2]
Age at Diagnosis	61 (53, 70)	61 (53, 70)	58 (50, 70)	64 (52, 72)	0.4
Sex					0.6
Female	618 (47%)	546 (48%)	31 (41%)	41 (48%)	
Male	687 (53%)	599 (52%)	44 (59%)	44 (52%)	
Race					
Asian-Far East/Indian Subcont	85 (6.5%)	71 (6.2%)	6 (8.0%)	8 (9.4%)	
Black Or African American	102 (7.8%)	88 (7.7%)	7 (9.3%)	7 (8.2%)	
Other	35 (2.7%)	28 (2.4%)	5 (6.7%)	2 (2.4%)	
Pt Refused To Answer	37 (2.8%)	32 (2.8%)	2 (2.7%)	3 (3.5%)	
Unknown	6 (0.5%)	6 (0.5%)	0 (0%)	0 (0%)	
White	1,040 (80%)	920 (80%)	55 (73%)	65 (76%)	
Highest Stage					>0.9
Stage 1-3	549 (42%)	483 (42%)	30 (40%)	36 (42%)	
Stage 4	756 (58%)	662 (58%)	45 (60%)	49 (58%)	
MSI Score	0.65 (0.16, 1.57)	0.63 (0.16, 1.54)	0.68 (0.17, 1.64)	0.87 (0.18, 2.39)	0.14
TMB (Non-Synonymous)	6.1 (4.3, 7.8)	5.9 (4.3, 7.8)	6.9 (3.9, 8.2)	7.4 (5.2, 9.8)	<0.001
TP53 Mutant, n (%)	1,032 (79%)	918 (80%)	60 (80%)	54 (64%)	0.001
APC Mutant, n (%)	1,002 (77%)	884 (77%)	60 (80%)	58 (68%)	0.13
KRAS Mutant, n (%)	595 (46%)	521 (46%)	32 (43%)	42 (49%)	0.7
SMAD4 Mutant, n (%)	219 (17%)	190 (17%)	12 (16%)	17 (20%)	0.7
PIK3CA Mutant, n (%)	207 (16%)	172 (15%)	13 (17%)	22 (26%)	0.028
TCF7L2 Mutant, n (%)	139 (11%)	117 (10%)	9 (12%)	13 (15%)	0.3
FBXW7 Mutant, n (%)	116 (8.9%)	91 (7.9%)	15 (20%)	10 (12%)	0.001
SOX9 Mutant, n (%)	110 (8.4%)	94 (8.2%)	7 (9.3%)	9 (11%)	0.7
BRAF Mutant, n (%)	101 (7.7%)	82 (7.2%)	3 (4.0%)	16 (19%)	<0.001
ARID1A Mutant, n (%)	85 (6.5%)	67 (5.9%)	3 (4.0%)	15 (18%)	<0.001

[1] Median (Q1, Q3); n (%)

[2] Kruskal-Wallis rank sum test; Pearson's Chi-squared test; NA; Fisher's exact test

Primary Analysis: Multivariable Cox PH Models

Chemotherapy Group

In the chemotherapy group, BRAF, SMAD4, PIK3CA, KRAS, and APC mutations were all significantly associated with OS. BRAF mutations had the strongest harmful effect (HR = 2.05, 95% CI: 1.57-2.67, $p < 0.001$), followed by SMAD4 (HR = 1.49, 95% CI: 1.22-1.81, $p < 0.001$), then PIK3CA (HR = 1.27, 95% CI: 1.03-1.56, $p = 0.027$), and finally KRAS (HR = 1.24, 95% CI: 1.05-1.45, $p = 0.010$). For a protective effect, APC mutations were associated with better OS (HR = 0.68, 95% CI: 0.56-0.82, $p < 0.001$). TP53, FBXW7, TCF7L2, ARID1A, and SOX9 mutations were not significantly associated with OS in this group.

Table 2. Cox PH Results: Chemotherapy Group (adjusted for age, sex, race, stage, logMSI, logTMB)

Gene	HR (95% CI)[1]	P-value
BRAF	**2.05 (1.57–2.67)**	**<0.001**
APC	**0.68 (0.56–0.82)**	**<0.001**
SMAD4	**1.48 (1.22–1.81)**	**<0.001**
KRAS	**1.24 (1.05–1.45)**	**0.01**
PIK3CA	**1.27 (1.03–1.56)**	**0.027**
TP53	0.84 (0.69–1.02)	0.084
TCF7L2	0.8 (0.61–1.05)	0.107
ARID1A	0.8 (0.53–1.2)	0.281
SOX9	0.89 (0.66–1.2)	0.449
FBXW7	0.92 (0.69–1.24)	0.587

[1] HR > 1 = worse OS in mutation carriers. * p < 0.05.

Chemotherapy and Targeted Agents; Other/ Investigational Group

In the chemotherapy and targeted agent group, KRAS (HR = 2.76, 95% CI: 1.25-6.10, p = 0.012), PIK3CA (HR = 2.94, 95% CI: 1.23-7.02, p = 0.016), and SMAD4 (HR = 2.73, 95% CI: 1.23-6.02, p = 0.013) mutations were associated with worse OS. In the Other/Investigational group, only BRAF mutations were associated with OS (HR = 2.67, 95% CI: 1.12-6.38, p = 0.027).

Gene x MoA Interaction Analysis

20 interaction terms were tested across all ten models (2 models per gene). After Benjamini-Hochberg FDR correction, no interaction terms met the pre-specified FDR p<0.01 significance threshold from the primary analysis. The most *nominally* significant interactions were PIK3CA mutations and the chemotherapy and targeted agent (HR = 0.46, p = 0.016, FDR p = 0.315), SMAD4 and chemotherapy and targeted agent (HR = 0.48, p = 0.032, FDR p = 0.319), and KRAS and other/ investigational (HR = 1.81, p = 0.049, FDR p = 0.323), none of which survived the multiple comparisons correction.

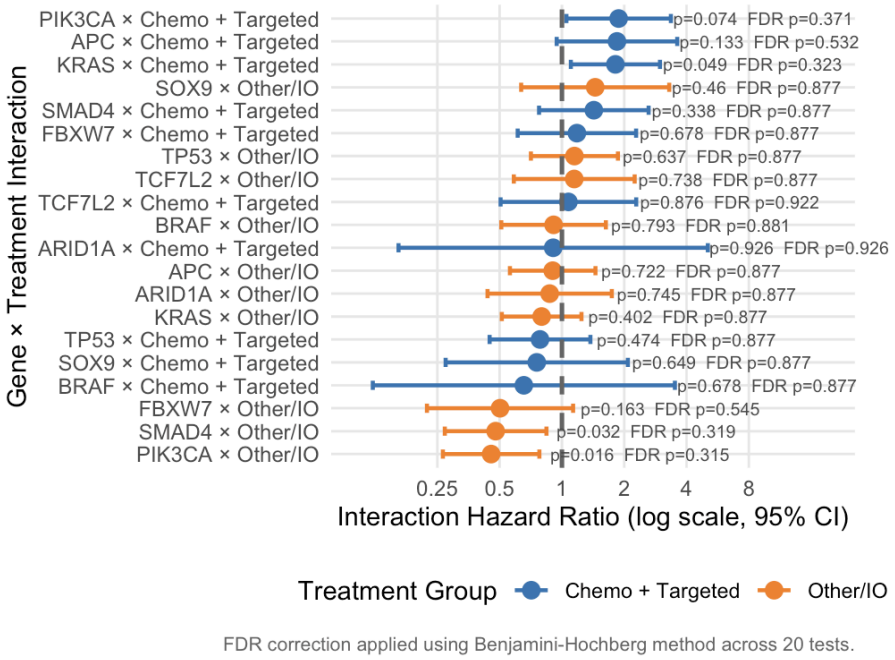

Figure 1. Forest plot of gene x treatment interaction HRs (reference group = chemotherapy)

PH Assumption

The PH assumption was evaluated using Schoenfeld residuals. All ten Cox models had significant global test statistics ($p < 0.001$). At the gene mutation level, APC ($p = 0.012$), ARID1A ($p = 0.019$), and BRAF ($p = 0.026$) showed significant violations. The PIK3CA x MoA interaction term showed a strong violation ($p = 0.00052$), and the BRAF x MoA interaction term showed a borderline violation ($p = 0.050$). These findings motivated secondary analyses for APC, BRAF, and PIK3CA.

Landmark Analysis

The landmark analysis split the cohort at median follow-up time, which was 35.5 months. After this analysis, one significant interaction term was found after FDR correction. The PIK3A mutation was associated with worse OS among patients on the chemotherapy and targeted treatment regimen in the later period (HR = 19.65, 95% CI: 4.14-93.28, $p = 0.0002$, FDR $p = 0.007$). Across all primary and secondary analyses, this interaction term was consistently significantly associated with worse OS after FDR corrections.

The KRAS mutation was *nominally* associated with the chemotherapy and target regimen in the later period (HR = 3.96, 95% CI: 1.44–10.88, $p = 0.008$), though this did not survive FDR correction (FDR $p = 0.154$). No significant interactions were identified in the early period

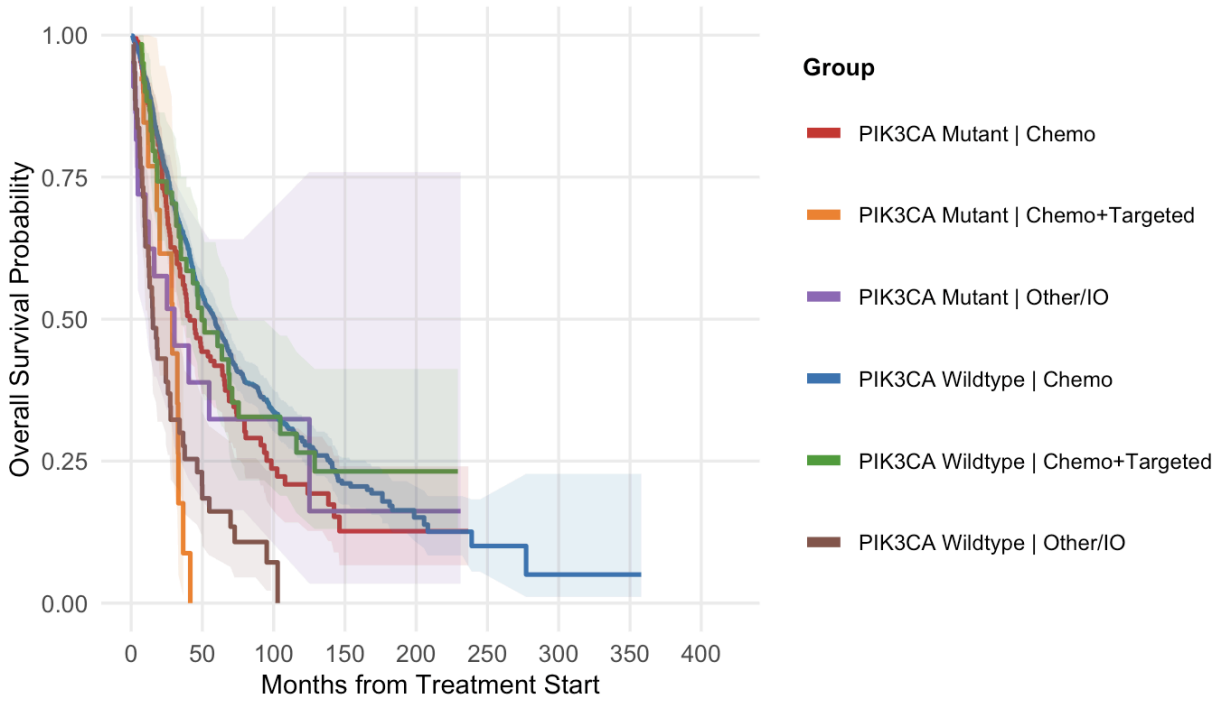

Figure 2. Kaplan-Meier Curves representing OS by PIK3CA mutation status and treatment

RMST Analysis and AFT analysis

In order to address the PH violations in the three identified genes of interest (APC, BRAF, and PIK3CA), a RMST analysis was carried out. These results confirmed findings from the Cox models for the BRAF and PIK3CA mutations in the chemotherapy group, where BRAF mutated patients on chemotherapy had significantly lower RMST compared to wildtype BRAF patients (difference = -28.99 months, $p < 0.001$, tau = 159.5 months). PIK3CA mutant patients followed the same direction (difference = -13.95 months, $p = 0.041$, tau = 212.9 months). No significant RMST differences were observed for BRAF or PIK3CA in the chemotherapy and targeted agent or other/ investigational groups.

The Weibull AFT models exhibited directional agreement with the Cox PH models for all tested interactions. The PIK3CA x Other/ Investigational interaction was significant in both (Cox $p = 0.016$, AFT $p = 0.005$), and PIK3CA x Chemo and Targeted showed a significant AFT p-value of 0.047, which was consistent with the landmark analysis results.

Table 3. RMST Analysis: Mutant v Wildtype by treatment group

Gene	Treatment	Tau (mo.)	RMST Diff (mo.)[1]	95% CI	P-value
APC					
APC	Chemo	249.4	10.92	16.73 to 5.1	0.123
APC	Chemo	249.4	10.92	16.73 to 5.1	0.123
APC	Chemo+Targeted	200.1	-29.27	123.19 to -181.74	0.237
APC	Chemo+Targeted	200.1	-29.27	123.19 to -181.74	0.237
APC	Other/IO	92.5	2.46	30.06 to -25.14	0.771
APC	Other/IO	92.5	2.46	30.06 to -25.14	0.771
PIK3CA					
PIK3CA	Chemo	212.9	-13.95	39.66 to -67.56	0.041
PIK3CA	Chemo+Targeted	37.3	-4.94	17.27 to -27.15	0.13
PIK3CA	Other/IO	92.5	13.44	23.47 to 3.4	0.156
BRAF					
BRAF	Chemo	159.5	-28.99	51.32 to -109.3	<0.001
BRAF	Chemo+Targeted	34.8	-6.24	41.62 to -54.1	0.501
BRAF	Other/IO	92.5	-0.39	40.51 to -41.3	0.97

[1] Diff = Mutant – Wildtype. Negative = worse OS in mutation carriers. Tau = 90% of minimum arm max follow-up.

Bootstrap Validation

To evaluate the robustness of preliminary and secondary analyses' results, bootstrap validation using 250 resamples was used. Only one gene x treatment interaction term remained robust after bootstrapping: PIK3CA x Chemo and Targeted. All other 95% confidence intervals computed crossed 1, which was the null value and consistent with non-significance after FDR corrections in the primary analysis of interaction terms.

Sensitivity Analysis

After removing the 7 outliers identified by deviance residuals, the interaction HR estimates changed by less than 10% for only 5 of interactions, which indicates stability despite inclusion of outliers. Three interactions with the Other/ Investigational group had a greater than 10% change: APC, PIK3CA, and SMAD4, which was expected due to the smaller sample size of this group.

Model Performance

The mean C-index across all 10 cox models fit in the primary analysis was 0.640 (range: 0.636 - 0.647). This indicates moderate discriminative ability. Martingale residual plots confirmed the correct functional form for continuous covariates, and led to the inclusion of quadratic terms for age, logMSI, and logTMB.

Discussion

These results contribute to emerging evidence that suggests somatic mutations carry meaningful information in metastatic CRC patients, particularly in the context of chemotherapy. The nominal associations identified in the chemotherapy group (BRAF, SMAD4, KRAS, PIK3CA, and APC) are consistent with existing literature. BRAF is a common genetic biomarker that adversely affects metastatic CRC patients.[4,5] The protective association observed for APC mutation is more nuanced. APC-mutant tumors may represent a WNT-driven disease subtype with somewhat more predictable biology than

tumors driven by different oncogenic mechanisms.[6,7] All together, the consistency of these findings across Cox, RMST, and AFT frameworks, despite PH violations, strengthens confidence that these associations reflect genuine biological signals rather than statistical noise.

The failure to uncover significant gene x treatment interactions after FDR correction in the primary analysis should not be interpreted as evidence that the interactions do not exist. With only 76 patients in the chemotherapy and targeted group and 85 in the other/ investigational group, this study was slightly underpowered to detect more nuanced interaction effects. The nominal signals for PIK3CA, SMAD4, and KRAS are understood as directions for future investigation rather than as null findings.

The landmark analysis finding of a PIK3CA x chemotherapy and targeted interaction in the late period is interesting, as it survived FDR correction and showed directional agreement with the AFT models and bootstrap validation. Biologically, this is possible given PIK3CA's role in the PI3K/AKT/mTOR pathway and its known associations with resistance to EGFR-targeted agents commonly used in the chemotherapy and targeted group.[8,9,10,11] However, the large point estimate and wide confidence interval are strong indicators of instability, possibly driven by sparse late events in a small subgroup, and this result should be treated as hypothesis-generating.

Several limitations exist in this study. The imbalance across treatment groups resulted in slightly underpowered analyses for the smaller treatment groups. This reflects real-world treatment distributions, but it highlights the need for larger datasets to be able to study the associations across various treatments groups. The observational design introduces residual confounding that covariate adjustment cannot fully address. The other/ investigational group is likely enriched for MSI-High patients who respond positively to immunotherapy *regardless* of other mutation status. Therefore, interaction effects may partially reflect MSI-driven selection rather than true gene x treatment effects. The binary mutation classification used here also obscures clinically important variances. The dataset did not provide variant-level distinctions such as KRAS G12D versus G12V, or BRAF V600E versus non-V600E, which carry biological implications that a simple mutant/ wildtype indicator cannot represent wholly. Finally, the trade-off between multiple testing correction and statistical power is expected in an exploratory interaction analysis at this level and the FDR correction applied is rigorous *but* conservative.

Conclusion

This study demonstrated the potential and constraints of using real-world clinicogenomic data to identify associations between mutations and OS by treatment groups in metastatic CRC. Across robust primary and secondary analyses, BRAF, PIK3CA, KRAS, and APC showed consistent relevance in the chemotherapy group, which aligns with existing literature. The PIK3CA x chemotherapy and targeted interaction in the late period showed the most promise for future study directions. Failure of interaction terms surviving the FDR correction reflect the conservative nature of FDR and the limited sample sizes in treatment groups. Future studies should prioritize external validation of the PIK3CA interaction, expansion to larger and more treatment-diverse cohorts, and incorporation of variant-level mutation data to more precisely capture the biological heterogeneity obscured by binary mutation indicators.

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